Does the Quantum Circuit Help? A Controlled Ablation of a Hybrid Quantum–Classical Protein Contact Predictor

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Abstract

Hybrid quantum–classical neural networks are frequently proposed for bioinformatics tasks, but reported gains are rarely attributed to the quantum component through a controlled experiment. We build a hybrid model that predicts inter-residue contact and distance maps from sequence alone, combining ESM-2 protein language model representations with a 12-qubit MERA variational circuit, and evaluate it on the official ProteinNet CASP11 validation set under the CASP assessment protocol. The system outperforms ESM-2's own native contact prediction across all three sequence-separation ranges, by 19.0, 15.8 and 2.1 percentage points. We then replace the quantum circuit with a classical linear layer of equivalent capacity, holding data, seed, architecture and training protocol fixed. Across two independent replicates the performance difference is -0.0003 and -0.0071 in long-range P@L/5, both marginally favouring the classical baseline. We attribute the observed improvement to the classical architecture rather than to the quantum ansatz, and trace the null result to the vanishing gradient of the circuit parameters and to the encoding bandwidth of angle-based state preparation.

1 Introduction

Predicting which residue pairs are in spatial contact from sequence alone is a long-standing problem in structural bioinformatics. The field advanced sharply when covariation signals extracted from multiple sequence alignments (MSAs) were combined with machine learning: the best group in CASP11 reached 27% precision on the top $L/5$ long-range contacts, against roughly 20% in the preceding edition [1, 2].

More recently, protein language models trained by masked-token prediction over hundreds of millions of sequences have been shown to encode structural information in their attention maps without any structural supervision [3], enabling contact prediction from a single sequence and removing the cost of building alignments at inference time.

Variational quantum circuits have been proposed as feature extractors for such tasks, motivated by the exponential dimension of Hilbert space and by the inductive bias of tensor-network ansätze, which were originally developed to represent quantum states with multi-scale entanglement structure. The appeal for protein contact prediction is intuitive: the problem exhibits correlations at several length scales, from local secondary structure to long-range tertiary packing.

Reported results, however, seldom isolate the contribution of the quantum component. A hybrid model that outperforms some baseline may owe its performance entirely to its classical parts, and without an equivalent-capacity classical control the attribution cannot be made. This is the question we address.

We make three contributions. First, we design the hybrid architecture so that the quantum route is *interchangeable* with a classical layer of matched capacity, enabling a controlled ablation. Second, we include the native contact prediction of the pretrained language model as an explicit reference, separating what the architecture adds from what the representations already provide. Third, we evaluate under the CASP protocol with exclusive separation ranges, making the figures directly comparable with the published record.

2 Method

2.1 Architecture

For a residue pair (i, j) we build a symmetric feature vector from ESM-2 `t12_35M` representations [3]:

$$\mathbf{x}_{ij} = \left[\, \mathbf{e}_i + \mathbf{e}_j \,\|\, |\mathbf{e}_i - \mathbf{e}_j| \,\|\, \tfrac{j-i}{L} \,\|\, \mathbf{a}_{ij} \,\right] \in \mathbb{R}^{1201}, \quad (1)$$

where $\mathbf{e}_i \in \mathbb{R}^{480}$ are per-residue embeddings and $\mathbf{a}_{ij} \in \mathbb{R}^{240}$ the pairwise attention maps.

A classical funnel compresses $\mathbf{x}_{ij}$ to $n = 12$ angles, which are encoded into the circuit by single-qubit rotations. The circuit is a MERA ansatz [4] with entangling blocks and disentangling gates across renormalisation levels, and data re-uploading [5]. Measurement yields 27 expectation values: $12\langle Z_i\rangle$, $9\langle Z_iZ_j\rangle$, $3\langle X_i\rangle$ and $3\langle Y_i\rangle$. These, concatenated with a residual copy of the input angles, feed classical layers and three prediction heads (distance bin and two orientation angles). Figure 1 summarises the pipeline.

2.2 Controlled ablation

The circuit maps $\mathbb{R}^{12} \to \mathbb{R}^{27}$, the same signature as a `Linear`$(12, 27)$ layer, with comparable parameter counts (312 versus 351). Swapping one for the other leaves the rest of the model untouched, so any performance difference is attributable to that single factor. Both arms are trained consecutively within one run from the same seed. The circuit is explicitly reinitialised in each arm: it is instantiated as a shared

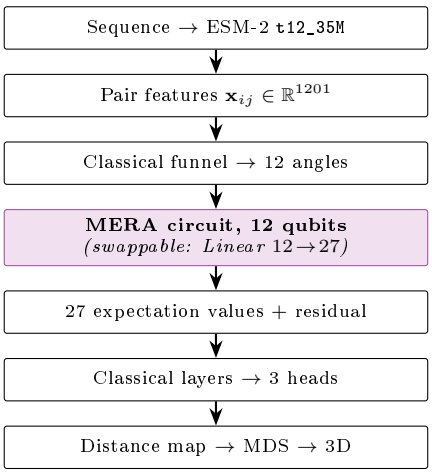


Figure 1: System pipeline. Only the highlighted block differs between the two ablation arms; every other component, the data and the random seed are held fixed.

global object and would otherwise inherit weights from the main pipeline.

3 Experimental setup

Training uses 15,000 proteins sampled uniformly from the `training_100` subset of ProteinNet CASP11 [6], yielding 200,000 residue pairs balanced across five distance bins. The split is by whole protein, so no pair from a given chain appears in both partitions. All reported metrics are computed on the 224 proteins of the official CASP11 validation set, unseen during training.

Training minimises a multitask cross-entropy over the distance bin and two orientation angles, with AdamW, gradient clipping and early stopping on the monitoring split. The quantum parameter group receives a separate, higher learning rate with a linear warmup, since its gradients are substantially smaller than those of the classical weights.

We follow the CASP contact assessment protocol: pairs are grouped into *exclusive* separation ranges — short ($6 \le |i-j| \le 11$), medium (12–23) and long (≥ 24) — and precision is measured on the top L, $L/2$, $L/5$ and $L/10$ ranked pairs. We additionally report Matthews correlation (MCC) and the Xd score, both part of the official assessment [1]. Long-range figures are the ones CASP quotes and the only ones directly comparable with the published record.

4 Results

4.1 The hybrid system improves on its input

Table 1 compares the trained system against two references: the contact density of each range, which is the expected performance of a random predictor, and the native contact prediction of ESM-2 itself.

The system improves on ESM-2 in all three ranges, and MCC, Xd and AUC-PR agree with precision, so the gain is not an artefact of a single metric. The margin narrows sharply with separation: in the long-range regime precision reaches 10.9%, only 1.8× the contact density, and an MCC of 0.018 indicates limited discriminative power. Three-dimensional reconstruction, obtained by multidimensional scaling of the predicted distance map followed by geometric refinement, is summarised in Table 2. The mean values are modest and, more tellingly, the standard deviation of GDT_TS is as large as its mean. The dispersion is strongly associated with chain length: GDT_TS exceeds 0.35 for chains below 100 residues and falls under 0.03 above 400. The explanation is geometric — reconstruction must satisfy all pairwise distances simultaneously, and their number grows quadratically with length, so prediction error compounds. In two of the thirty proteins the reconstruction collapses entirely (GDT_TS $= 0$, RMSD > 24 Å); we report these degenerate cases rather than excluding them.

Table 1: Contact precision on the 224-protein CASP11 validation set, by exclusive separation range.

Range	Model	P@L	P@L/5	MCC
Short	This work	**87.5**	**88.9**	0.101
	ESM-2 native	69.7	69.9	0.018
	Random (density)	65.5		—
Medium	This work	**38.9**	**40.8**	0.071
	ESM-2 native	25.0	25.0	0.009
	Random (density)	22.4		—
Long	This work	**10.2**	**10.9**	0.018
	ESM-2 native	8.6	8.9	0.007
	Random (density)	6.2		—

Table 2: Three-dimensional reconstruction over 30 validation proteins.

Metric	Mean	Std.	Best
RMSD (Å)	14.24	—	3.55
GDT_TS	0.114	0.111	0.485
GDT_HA	0.028	—	—
TM-score	0.222	0.094	0.425
lDDT	0.363	—	0.542

4.2 The quantum circuit does not contribute

Table 3 reports the ablation. The difference is three ten-thousandths in the first replicate and seven thousandths in the second, both favouring the classical layer and both far below the between-protein variability (the standard deviation of TM-score alone is 0.094). The classical baseline is remarkably reproducible across replicates (0.1050 and 0.1051) whereas the circuit is more dispersed (0.1047 and 0.0980).

Crucially, this null result is obtained in a regime where *both* arms work: each of them beats the ESM-2 reference in all three ranges. We are not comparing two failing systems but two functioning ones that are indistinguishable.

Table 3: Ablation of the quantum route. Long-range P@L/5; two independent replicates differing only in the ProteinNet subset used for training.

Replicate	MERA	Linear	Δ
1 (`training_30`)	0.1047	0.1050	−0.0003
2 (`training_100`)	0.0980	0.1051	−0.0071

Table 4: Long-range P@L/5 in context. CASP figures refer to free-modelling domains, the hardest category.

System	Approach	P@L/5
CONSIP2 (CASP11 best)	MSA + covariation	27.0%
CASP10 best	no covariation	∼20%
This work	**no MSA**	**10.9%**
ESM-2 native	no MSA	8.9%
Random predictor	—	6.2%

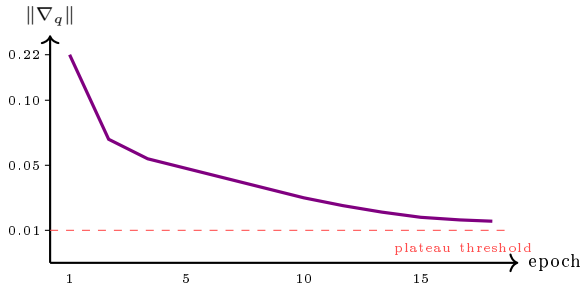


Figure 2: Gradient norm of the quantum parameters during training. The sustained decay towards the plateau threshold indicates that the circuit stops receiving learning signal, which is consistent with the equivalence observed in Table 3.

4.3 Context within CASP11

Table 4 places the result within the edition from which our validation set is drawn. Twenty-nine groups took part in the CASP11 contact category, at least eight of them using evolutionary coupling techniques [1]. The leader, CONSIP2, was clearly separated from the rest — CASP11 was the edition in which covariation methods demonstrated their superiority — while methods without covariation analysis lagged substantially behind.

5 Discussion

Figure 2 offers a mechanistic account. The gradient norm of the quantum parameters decays steadily from 0.22 to about 0.015, approaching the threshold below which the circuit effectively stops receiving learning signal. A circuit whose parameters barely update converges to an essentially fixed transformation, functionally equivalent to the linear layer that replaces it. This is consistent with the barren plateau phenomenon reported for variational circuits [7].

A structural limitation bounds what any ansatz can contribute here. Angle encoding injects one value per qubit and measurement extracts 27 observables: although 12 qubits span a 4096-dimensional Hilbert space, the data occupy a negligible fraction of it. The bottleneck is the encoding and readout bandwidth, not the internal structure of the circuit. This suggests that exploring alternative ansätze, without changing the encoding scheme, would not alter the outcome — consistent with the absence of measurable differences we observe, and with recent work identifying encoding selection as the central problem in quantum kernel methods [8].

We further verified that the ceiling is not one of data or of model capacity, by ruling out four alternative explanations experimentally. *(i)* Scaling the training set from 15,000 to 200,000 pairs and from 1,200 to 15,000 proteins removed the initial overfitting — validation loss no longer diverges — but left validation accuracy essentially unchanged at 0.35. *(ii)* Training on `training_100` instead of `training_30`, which quadruples the source pool and adds substantial sequence redundancy, produced differences within noise; the coevolutionary signal one might expect from redundant homologues is not recovered by this system. *(iii)* Rebalancing the sampler by separation range proved counterproductive: measurement showed that random sampling per distance bin already yields 67% long-range pairs by combinatorics, which the rebalancing reduced to 33%. *(iv)* Redesigning the model to predict the full $L \times L$ map with a dilated residual CNN, whose receptive field spans about 180 residues and can therefore exploit triangle consistency and the diagonal patterns of beta sheets, improved short and medium range but not the long-range regime.

Relative to the published record, 10.9% sits below the 27% of the best CASP11 group and below most participants using covariation. The gap is expected: those methods extract coevolution from alignments of tens to hundreds of homologues, which is precisely the signal that resolves contacts between distant regions of the chain. Our system operates on a single sequence, relying on whatever structural information the language model internalised during pretraining, and the 35M-parameter variant used here bounds how much that can be.

6 Conclusion

We presented a hybrid quantum–classical contact predictor and, more importantly, a controlled experiment to determine what its quantum component contributes. The system improves on the native prediction of the underlying language model across all separation ranges, but replacing the 12-qubit MERA circuit with a capacity-matched linear layer leaves performance unchanged across two independent replicates. The improvement is therefore attributable to the classical architecture. We trace the null result to a vanishing quantum gradient and to the encoding bandwidth of the angle-based scheme, and we argue that progress in this setting requires richer input representations and alternative encodings rather than deeper or more ex-

pressive ansätze.

Code and data preparation scripts are available at https://github.com/marrocrod/qnn-protein_structure-prediction.

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References

[1] B. Monastyrskyy *et al.*, "New encouraging developments in contact prediction: Assessment of the CASP11 results," *Proteins*, vol. 84, pp. 131–144, 2016.

[2] D. T. Jones *et al.*, "Accurate contact predictions using covariation techniques and machine learning," *Proteins*, vol. 84, pp. 76–86, 2015.

[3] Z. Lin *et al.*, "Evolutionary-scale prediction of atomic-level protein structure with a language model," *Science*, vol. 379, pp. 1123–1130, 2023.

[4] G. Vidal, "Entanglement renormalization," *Phys. Rev. Lett.*, vol. 99, 220405, 2007.

[5] A. Pérez-Salinas *et al.*, "Data re-uploading for a universal quantum classifier," *Quantum*, vol. 4, p. 226, 2020.

[6] M. AlQuraishi, "ProteinNet: a standardized data set for machine learning of protein structure," *BMC Bioinformatics*, vol. 20, 311, 2019.

[7] J. R. McClean *et al.*, "Barren plateaus in quantum neural network training landscapes," *Nat. Commun.*, vol. 9, 4812, 2018.

[8] D. D. Tung *et al.*, "Qmes: Quantum meta-learning for encoding selection in quantum kernel methods," *arXiv:2609.04652*, 2026.